

LUMAFLOW · REQUIREMENTS

LumaFlow requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-01	Amish Chadha	Rerun for the constructable design (LMF-DDR-003); R16 reported against the value-engineering target; R15 note
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish; R16 target \$340, now met (LMF-CAL-001 v0.3)
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R1 to 1.2 L/min, R8 restated, R10 adds the thermal cut-back, new R17 pressure limiter; status from LMF-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Record Amish's decisions (LMF-DDR-001); R16 redefined to \$225; R7 design stress sourced (6.8 MPa); status from LMF-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

LumaFlow requirements

These are the requirements for the concept, checked by calculation at TRL 3 in LMF-CAL-001 v0.4, for the constructable design of LMF-DDR-003. The status column reports that note; bracketed tags such as [B5] point to the line of `docs/04-calcs/sizing.py` that carries the number. One requirement is **not met**: R3 (dose at the 70 %/cm test condition, accepted as not met by decision D1). R16 (cost) is reported against its value-engineering target: the constructable design is estimated at USD 402, USD 62 over the USD 340 target. One is at risk: R4. The dose targets (R2, R3) and the budget (R16) follow Amish's decisions in LMF-DDR-001; R1, R8, R10 and the new R17 follow the recommendations Amish accepted on 2026-09-25 (LMF-DDR-002).

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Status (LMF-CAL-001 v0.2)	Verification
R1	Deliver a useful drinking-water flow	1.2 L/min (0.32 gpm) at the faucet, capped by a flow restrictor (was 2.0 L/min; reduced by LMF-DDR-002 so that R2 is met with margin)	Met by design [A3]	Restrictor rating; flow calculation
R2	Deliver the Class A dose level in clear water	Reduction equivalent dose (RED) of 40 mJ/cm ² or more at 1.2 L/min and UV transmittance (UVT) of 90 %/cm	Met (51.9 laminar to 76.3 plug mJ/cm ²) [B5]	Ray-trace dose calculation; later biodosimetry
R3	Deliver the Class A dose level at the NSF/ANSI 55 test water quality	RED of 40 mJ/cm ² or more at 1.2 L/min and 70 %/cm UVT	Not met (17.4 to 21.8 mJ/cm ²) [B6]; kept visibly not met by decision D1	Dose calculation
R4	Monitor the dose and fail safe	UV-C sensor in the tube wall reading through the water; visual and audible alarm and outlet valve closed within 1 s when the estimated dose falls below 40 mJ/cm ² , the sensor fails or power is lost	At risk: the wall signal is 16.3 nA at 90 %/cm and 0.38 nA at 70 %/cm, and the proportional mapping trips below about 89 %/cm, but the mapping and 1 s fail-safe need firmware and a test [C1, C4]	Signal calculation; later firmware and test
R5	Switch with the flow	LEDs on within 0.5 s of flow above 0.3 L/min; LEDs held on for 5 s after flow stops; LEDs off otherwise	Met (0.24 s worst case with a 15 Hz per L/min sensor) [D2]	Timing calculation; later firmware sketch review
R6	No mercury and no warm-up	No mercury in any part; full UV output within 0.1 s of switch-on	Met by design	Design review; LED datasheet
R7	Withstand household water pressure	Working pressure 8 bar (116 psi); quartz window stress 6.8 MPa (1,000 psi) or less at 8 bar	Met (6.18 MPa with a 10 mm window on a 51 mm seat) [G4]	Plate stress calculation
R8	Low pressure drop	0.5 bar (7 psi) or less across the unit at 1.2 L/min, excluding the flow restrictor, which absorbs excess supply pressure by design; sensor and valve flow coefficients from their datasheets	Met (0.15 bar on assumed flow coefficients; datasheet values to be confirmed when parts are chosen) [G2]	Component data and hydraulic calculation
R9	Low energy use	25 W or less while water flows; 0.5 W or less on standby	Met (22.6 W and 0.21 W) [E1, E2]	Power budget calculation
R10	Keep the LEDs cool	LED board 50 °C or less with 30 °C cabinet air and 25 °C water, including the 5 s run-on; a board temperature sensor cuts LED current above 50 °C	Met (37.1 to 44.8 °C over the film-coefficient range) [F3]	Thermal calculation; cut-back reviewed as a firmware rule
R11	Food-safe wetted parts	Every wetted material food-contact grade (316 stainless steel, PTFE, fused quartz, EPDM or silicone O-rings, acetal); no UV-C on any plastic other than PTFE	Met by design (316 lower cap; PTFE liner, top disc and 316 insert shield the acetal upper cap)	Material review of the model
R12	Contain the UV-C light		Met by design	

ID	Requirement	Target	Status (LMF-CAL-001 v0.2)	Verification
		No UV-C outside the unit in normal use; LEDs cannot run with the LED head or enclosure lid removed		Design review of interlock and light path
R13	Electrical safety	Only 24 V DC at the unit; mains confined to a certified external adapter; fuse on the 24 V input	Met by design	Design review
R14	Fit under a kitchen sink	Envelope 350 x 150 x 350 mm (13.8 x 5.9 x 13.8 in) or less, excluding the adapter	Met (277 x 100 x 322 mm) [H1]	Parametric model
R15	Easy service	Window and LED head replaceable with hand tools in 15 min or less; status LED shows run, fault and service due	Not verifiable at TRL 3 (LED head off on four screws from below and window out after six ring screws, with no plumbing disturbed; needs a build to time)	Timed service on a build
R16	Low cost and buildable	Value-engineering target of USD 340 per unit in parts (budget_usd , a hypothetical control target, not a limit; set from \$200 by decision D2, then \$225, then \$340 by the top-up Amish approved on 2026-09-26); no custom PCB for the first build	Over the value-engineering target by USD 62 (estimated cost of the constructable design USD 402; the concept was \$338.00) [I1]	Priced BOM
R17	Pressure protection (installation)	A pressure-limiting valve upstream of the unit, set at 4 bar (58 psi) or less, so that transients stay within the 8 bar window rating (new, LMF-DDR-002)	Met (window 3.09 MPa at 4 bar, 6.18 MPa at a transient of twice the setting) [G5]	Plate stress calculation; installation instructions

■ Assumptions

- **Water quality.** The design water is clear, pre-filtered well water (decision D9): UVT of 90 %/cm or better at 275 nm, turbidity below 1 NTU, after a 5 µm sediment pre-filter. UVT must be measured for each site. Iron, manganese and hardness foul the window and must be within the limits commonly given for UV systems; this is an installation requirement, not a LumaFlow feature.
- **Dose target.** 40 mJ/cm² is the NSF/ANSI 55 Class A dose level. Reaching it would not make LumaFlow Class A compliant or certified. It inactivates bacteria and protozoa such as *Cryptosporidium* by several logs but gives a much smaller log reduction of adenovirus, which the US EPA dose tables show needs about 186 mJ/cm² for 4 logs (EPA UV toolkit). RED in LMF-CAL-001 is computed for a challenge organism of 20 mJ/cm² per log.
- **Liner reflectance.** R2 rests on a wetted reflectance of 0.95 for the high-reflectance PTFE liner, which must be measured; plain PTFE (0.80) would give 46.3 mJ/cm², still above 40 (LMF-CAL-001, B8).
- **Pressure.** The pressure-limiting valve (R17) is part of the installation, like the sediment pre-filter, and is not in the LumaFlow BOM.
- **Demand.** 10 to 20 L per household per day, drawn in about 20 short draws.
- R3 is kept as stated so that the gap stays visible. The R16 figure of \$340 is a value-engineering target (Amish, 2026-10-01); the cost drivers and savings worth trying are in the design decisions register, LMF-DEC-001.